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Adaptive management of ecological systems under partial observability

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ABSTRACT

Adaptive management has a long history in ecology and conservation. Uncertainty in both the state of a system and the model defining its dynamics are fundamental challenges in adaptive management of complex ecological systems. Traditional approaches in conservation biology often ignore one or both sources of uncertainty due to the computational complexity involved. Here, we show that underestimating the role of uncertainty in both model estimation and decision-making results in aggressive decision rules which can potentially lead to the dramatic decline and possible collapse of a population, species, or ecosystem. We propose an approximate solution to adaptive management of ecological systems under both model and state uncertainties that is computationally feasible and applicable to complex management problems and provide a software for detailed implementation of our method, http://doi.org/10.5281/zenodo.1161521. We apply the proposed method in a marine ecosystem management context and show that by learning from historical data and arrival of new observations, decision makers can adapt their policies to avoid decline in the population and reach a sustainable population stability.

1. Introduction

Many studies in natural resource management emphasize the importance of incorporating uncertainty in the management process through adaptive management (Walters and Hilborn, 1978; Humphrey and Stith, 1990; Lancia et al., 1996; Williams, 2011a). In general, adaptive management is defined as a management that learns by doing (i.e. planning) and adopts management policies to reflect new observations (i.e. learning) (Walters and Holling, 1990; Williams and Brown, 2016). Adaptive management has a long history in ecological literature and have been widely used in behavioral ecology (Mangel and Clark, 1988), optimal harvesting in fisheries (Walters and Hilborn, 1976; Reed, 1979), as well as conservation biology and natural resource economics (Mangel, 1985; Moore et al., 2010; Britton et al., 2011). Walters and Hilborn (1976) are among the first implementations of adaptive management to control the population of Fraser River sockeye salmon under uncertainty in models describing the population dynamics.

Less computationally intensive approaches for natural resource management have since followed, such as management strategy evaluation (MSE) (Smith, 1994; Mapstone et al., 2008; Bunnefeld et al., 2011). Instead of attempting to solve for the optimal management policy in the space of all possible sequences of actions a manager could take, MSE evaluates consequences of a pre-determined set of strategies defined based on management objectives and provides the outcome of those simulations to the decision maker. Although this approach has proved to be promising in many applications in natural resource management, it is hard to quantify the quality of the selected management strategies compared to the optimal management.

Optimal management is the one strategy that results in the best outcome among all possible strategies and hence can be found by optimizing the management objectives over the life-span of the natural resource. Methods from the domains of decision theory and optimal control (Bertsekas, 1996; Sutton and Barto, 1998), such as Markov decision process (MDP), can be used to determine such an optimal policy, which has been a basis of adaptive management theory in ecology for years (Reed, 1979; Mangel, 1985; Clark and Kirkwood, 1986; Mangel and Clark, 1988; Sethi et al., 2005).

One of the main limitations of Markov decision process is the strong assumption about full observability of the system's state. For example, in fisheries management, the assumption is that the population biomass of the fishery of interest (i.e. state of the system) is measured without error, and fisher has certain knowledge of the exact population size before setting the harvest quota each year (i.e. management action). Relying on this assumption, Reed (1979) proves that the intuitive results of deterministic models (Beverton and Holt, 1957; Schaefer, 1957) remain optimal for maximizing the expected economic return of fisheries subject to stochastic growth. Clark and Kirkwood (1986) acknowledge the importance of the state uncertainty, due to error in the measurements, to the optimal ecological management, while observing

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that if this assumption were relaxed, the "difficulty of the problem increases markedly". Due to this difficulty and the conclusions of Reed (1979), most of the literature has focused on adaptive management with full observability of state space (e.g. certain knowledge of population biomass) and sidesteps this complexity in the decision making step of adaptive management (e.g. setting harvest rules) (Costello et al., 2016; Britten et al., 2017).

This has long begged the question: if the stochastic growth can be safely ignored in determining the optimal strategies for controlling the population (Reed, 1979), is the same true for measurement error? Since then, researchers have proposed approximate methods to incorporate the effect of measurement error on the optimal management policy and have come to a counter-intuitive conclusion that in presence of measurement error, the resulting policies are less conservative than the deterministic solutions (Ludwig and Walters, 1981; Clark and Kirkwood, 1986; Roughgarden and Smith, 1996; Engen et al., 1997; Sethi et al., 2005). Clark and Kirkwood (1986) acknowledge these counter-intuitive results: "results appear to contradict the conventional wisdom of renewable resource management, under which high uncertainty would call for increased caution in the setting of quotas". We show that the reason for this counter-intuitive result arises from non-trivial assumptions made to simplify the decision optimization process of management and to get past the computational complexity. By fully incorporating the effect of measurement error, we illustrate that the resulting optimal strategies for controlling the system is significantly different (and not necessarily less conservative) with respect to the deterministic solutions, and ignoring this effect can result in dramatic declines in the species population and possibly total collapse of the ecosystem.

Aside from the uncertainty in the estimating population biomass due to the measurement error, another challenge in adaptive management is uncertainty in the models defining dynamics of the population (i.e. model uncertainty). Walters and Hilborn (1978) provide an early non-mathematical overview that explores the effect of model uncertainty on management outcomes in natural resource management. They notice that once model uncertainty is introduced, finding the optimal management becomes very complex and hence solving the decision optimization under model uncertainty becomes intractable. This problem is known as the curse of dimensionality (Bertsekas, 1996) in decision theory. As a result, most of attempts to include model uncertainty in adaptive management of natural resources have focused on simplified examples (McDonal-Madden et al., 2010; Runge, 2011).

Here, we propose an approximate solution to adaptive management of complex ecological systems under both model and state uncertainties that is computationally scalable and applicable to the complex real-world management problems, which we call PLUS (Planning and Learning for Uncertain Systems). PLUS is comprised of two steps: model estimation (i.e. learning) and decision optimization (i.e. planning). We also provide a publicly available software and detailed implementation of our method, as an R package at http://doi.org/10.5281/zenodo.1161521. The proposed method builds upon recent advancements in adaptive management in domains of engineering and computer sciences (Doshi-Velez et al., 2012; Memarzadeh et al., 2014) and addresses several gaps in the literature regarding adaptive management and its application to natural resource management such as: (1) proposed adaptive management approaches being prone to curse of dimensionality and as a result are only applied to simplified examples, (2) proposed approaches make non-trivial assumptions and/or approximations to get past the computational barrier in the cost of losing flexibility and generality, which might also result in finding bad decision rules that can potentially lead to dramatic decline in population of biological species and possibly total collapse of the ecosystem.

We build the decision optimization step based on the framework of partially observable Markov decision process (POMDP) (Smallwood and Sondik, 1973; Sondik, 1978). POMDPs overcome one of the main limitations of Markov decision process (MDP, that are very popular in adaptive management of natural resources) and incorporate measurement error in the decision making by allowing partial observability of the state space through the introduction of belief state: manager's knowledge about system's state in a shape of probability distribution. This means that POMDP incorporates uncertainty in estimating the population biomass in the probabilistic manner. The reader should note that the belief state is well-known in the family of state space models for estimating the dynamics model under state uncertainty (Costello et al., 2016; Britten et al., 2017), although has been ignored and marginalized out in the decision making step. POMDPs are able to incorporate state uncertainty in decision making step with introduction of much higher computational complexity. Detailed formulation of POMDPs is reported in Appendix A. POMDPs have recently been used widely in ecological literature for control of animal population (Runge, 2013), natural resource management (Williams, 2009, 2011b), fishery management (Kling et al., 2017), and conservation in the face of climate change (Conroy et al., 2011). Although these studies emphasize partial observability and environmental variability as the challenging sources of uncertainty in the management of natural resources, they do not specifically propose an efficient method and software to get past the computational barrier. Moreover, POMDPs still require perfect knowledge of the models defining the system's dynamics, and as a result cannot incorporate model uncertainty. The approach taken by Walters and Hilborn (1976) of including the model parameters into the state space can be also implemented in POMDPs, however finding optimal strategies under such formulation is intractable and suffers from the curse of dimensionality. Here, we develop a heuristic based on minimization of the Bayes risk to overcome the computational complexity of decision making under model uncertainty and allow incorporation of both state and model uncertainties feasibly. The detailed formulation of the heuristic is reported in Appendix B. Overall goal of the decision optimization step is to find a decision (i.e. catch quota in the beginning of each year) that maximizes the long-term expected economic return of managing the system (i.e. fishery).

As mentioned before, the second step corresponds to estimating the dynamics model from historical observations and reducing the model uncertainty. In this step, the goal is to represent the historical observations in a shape of probability distribution over population dynamics model, and adaptively update the model and reduce the uncertainty once a new set of observations arrive in the following years. One of the main challenges in this step is that observations are only noisy measurements of the state of the system and hence need careful consideration. In general, one might not be able to represent the posterior distribution of the model parameters given the historical observations in closed-form, and as a result, need to adapt an approximate learning scheme based on Markov chain Monte Carlo (MacKay, 2003; Memarzadeh et al., 2014, Memarzadeh et al., 2016; Costello et al., 2016; Britten et al., 2017). Although in this article, we assume that the uncertainty is among a set of candidate models (as in Walters and Hilborn, 1976) and as a result, learning can be done in closed-form using Bayes rule (refer to Appendix C for detailed formulations).

2. Model formulation

We model the population dynamics of the fishery as follow,

$$x_{t+1} = f(x_t) - c_t + \sigma_t^X \tag{1}$$

where $x_t \in X$ denotes the population biomass, $c_t \in C$ is the harvested biomass (i.e. catch quota), with subscript t denoting years, σ_t^X is the annual biomass deviation due to stochasticity in the population dynamics (which we call growth noise), and f is the function governing the dynamics. We consider two candidate functions for f as the Ricker model (Ricker, 1954),

$$f(x_t) = x_t \exp\left(r\left(1 - \frac{x_t}{K}\right)\right) \tag{2}$$

and the Beverton-Holt model (Beverton and Holt, 1957),

$$f(x_t) = x_t\left(\frac{1+r}{1+\frac{rx_t}{K}}\right) \tag{3}$$

where K and r are the unknown parameters describing the carrying capacity and the intrinsic productivity of the population. It should be noted that these specific models are being chosen in order to be inline with the real-world database used later, and comparison to the literature that have used same models (Walters and Hilborn, 1976; Sethi et al., 2005), although the method can be directly extended and applied to more complex models such as age-structure models (Holden and Conrad, 2015).

As mentioned before, due to error in the measurements, population biomass cannot be identified with certainty. to model this uncertainty, we model the observations, z_t, as follows,

$$z_t = \epsilon_t^m x_t \tag{4}$$

where ϵ_t^m is the error in the measurements distributed log-normally, $\epsilon_t^m \sim \text{LogN}(x_t, \sigma_m)$. Other choices of noise other than log-normal can be easily used.

3. Results

To evaluate the effectiveness of learning different parameters in the population dynamics model (i.e. r, K, σ_t^X, and σ_m), we run 100 sets of independent simulations where one parameter is uncertain and others are fixed (Fig. 1). We compare the performance of our proposed method (i.e. PLUS) with the popular method in adaptive management of natural resources, Markov decision process (MDP), which ignores error in the measurements. The metric we use to quantify the learning process is the expected error in approximating the system's dynamics, that measures how far the manager's knowledge is from the actual model of population dynamics (The exact calculation of the expected error is reported in Appendix D); larger expected error means that manager's knowledge of the dynamics model is far away from the true model. The environment is modeled as partially observable and as a result MDP fails in convergence, while PLUS eventually learns the correct model parameters. It can be noted that learning is not efficient for the growth noise, σ_t^X. The reason for this roots in the work of Reed (1979), which proves that the optimal harvesting policy is not sensitive to the stochastic nature of the growth in the population and as a result, PLUS cannot fully identify the true parameter. Similar trends is shown for cases where population dynamics are governed by the Beverton-Holt model (Fig. S1 in Appendix E).

To emphasize the importance of adaptive management, we show that consecutive effects of ignoring the uncertainty in model parameters (e.g. fixing the parameters according to historical data that can be easily overestimated or underestimated) or ignoring the benefit of learning from new observations (denoted as planning) can result in significant decline in the economic return, and even total collapse of the system (Fig. 2). Specifically, we show through an example that overestimating the intrinsic productivity of the population, i.e. parameter r in the population dynamics model, can result in dramatic decline of the population and eventually the total collapse, while also underestimating r or ignoring the benefit of the future learning would result in the 20–35 % of economic loss. Of course, the effect would be more significant in complex models such as age-structure models, where there are more parameters playing a role in the dynamics of the population. We show that adaptive management can easily overcome this challenge by incorporating the uncertainty in management properly and reducing this uncertainty by learning from arrival of new observations. It is important to note that the significance of learning process can be variable according to: (1) how accurate the measurements are (direct relation to measurement error, σ_m) and this can be quantified as a pre-posterior analysis (Fig. S2 in Appendix E), and (2) how similar the candidate models are in our prior knowledge (Fig. S3 in Appendix E), higher similarity among candidates, harder to learn and identify the true model.

3.1. Application to real-world fishery management

We further evaluate the performance of the proposed adaptive management method (i.e. PLUS) on a real-world example of a marine ecosystem of Argentine hake in Patagonian Shelf and compare the adaptive management to the actual management implemented. The population biomass and harvest data are obtained from the RAM Legacy Stock Assessment database (Ricard et al., 2011). We have used the historical data to identify 144 candidate models with uncertainty in the parameters r, K, σ_t^X, and σ_m using maximum likelihood estimation through NIMBLE (NIMBLE Development Team, 2016). We show that adaptive management with PLUS adapts the harvesting policy as the new observations arrive about the population biomass each year and results in keeping the population in a sustainable level compared to the significant decline as a result of aggressive policies used in practice (Fig. 3). Specifically, the actual management does not react to the decline in the population biomass, which eventually resulted in a dramatic reduction in the population biomass by the year 2000, while PLUS is able to predict such decline and adapts a more conservative harvest policy after 1992 to avoid further decline and recovers the population to a stable level (twice the size of the actual population). Overall, adaptive management through PLUS results in a higher economic return in long-term and keeps the population in a sustainable level.

4. Discussion

Adaptive management is a crucial approach in solving many complex problems in natural resource management. It provides a framework for incorporating several sources of uncertainty in management such as state estimation uncertainty due to measurement error, uncertainty in the model describing dynamics of the ecosystem, and uncertainty in environmental variability. The two main steps in adaptive management correspond to decision making under uncertainty and reducing uncertainty by learning from arrival of new data. Most of the literature in adaptive management has focused on improving the second step of learning from data by developing methods based on Markov chain Monte Carlo and hierarchical Bayesian approaches (Costello et al., 2016; Britten et al., 2017), while sidestepping the importance of incorporating the uncertainty in the decision-making step. The reason for this, dates back to work of Reed (1979), which proved that the optimal decision rules of deterministic models are optimal under stochastic population dynamics. Since then, researchers have questioned this conclusion in presence of the measurement error in estimation of the population, although they have come to the conclusion that the resulting policies are more aggressive than the deterministic solutions (Ludwig and Walters, 1981; Clark and Kirkwood, 1986; Roughgarden and Smith, 1996; Engen et al., 1997; Sethi et al., 2005). The reason for these counter-intuitive conclusions is due to making nontrivial assumptions to get past the computational complexity of adaptive management under both model and state uncertainties. Here we emphasize the importance of including both model (i.e. structural) and state uncertainties in adaptive management and show that underestimating or ignoring uncertainties in state and/or models can potentially result in catastrophic consequences for the ecosystem.

The general framework of adaptive management has been applied to several problems in natural resource management including slowing rates of species extinction (Chades et al., 2011, 2012; Lampert et al., 2014), protecting natural resources against invasive species (Moore et al., 2010; Britton et al., 2011; Rout et al., 2011) and over-exploitation of resources such as fisheries (Walters and Hilborn, 1976; Fackler and Pacifici, 2014; Costello et al., 2016; Britten et al., 2017; Hastings et al.,

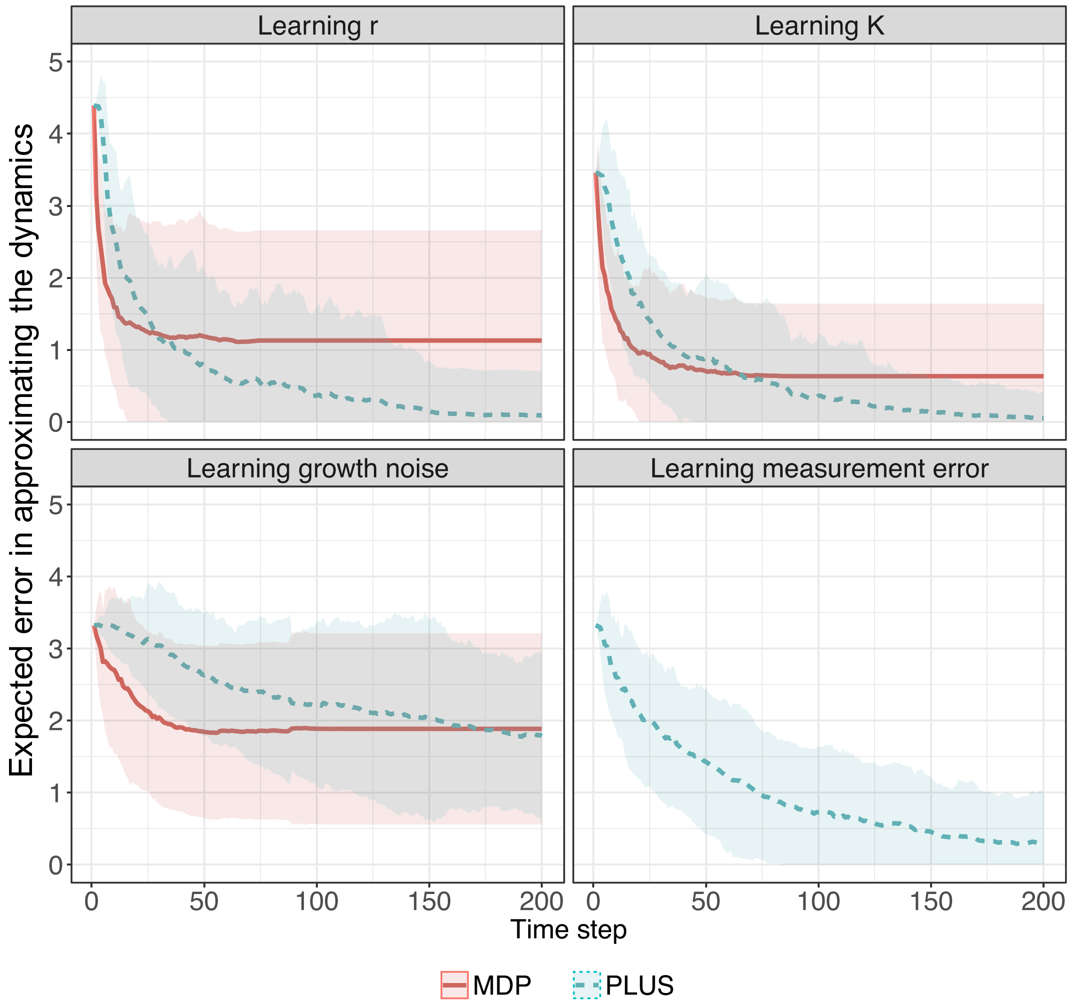

Fig. 1. This figure shows the mean and $+/-$ standard deviation of the expected error in approximating the true model describing the dynamics of the ecosystem by our proposed method (PLUS) and Markov decision process (MDP). Results are based on 100 independent simulations and learning occurs after arrival of each new data. Candidate models for different parameters are 21 models for $r \in \{0.5, 1.5\}$, 11 models for $K \in \{0.5, 1\}$, and 10 models for $\sigma_t^X = \sigma_m \in \{0.05, 0.5\}$. Population dynamics is governed by the Ricker model, and the true model for each case is $r = 0.95$, $K = 0.75$, and $\sigma_t^X = \sigma_m = 0.25$. The measurement error in the process for learning r, K, and σ_t^X is assumed to be $\sigma_m = 0.5$.

2017). Although we illustrate the application of our proposed adaptive management approach here to fisheries examples, this approach can be adopted to solve similar issues in other ecological problems such as bird species population (Chades et al., 2012) as well as management of endangered species against invasion (Moore et al., 2010; Britton et al., 2011). For example, Moore et al. (2010) study the impact of investment in quarantine to reduce the risk of pest invasion and surveillance to eradicate it before it gets out of control. They adopt the method of Markov decision processes (MDP) that we have discussed in this article, to model an island under the threat of invasion by a pest, and to optimally allocate management resources between quarantine and surveillance. This study assumes full observability of the state space (which represents the status of the pest), and complete certainty about model describing the population expansion of the pest. Similarly, Chades et al. (2012) sidestep the uncertainty in state estimation of managing population dynamics of the threatened Gouldian finch, a bird species endemic to Northern Australia, while including the uncertainty in population dynamics. It is worth to see how the proposed adaptive management method here, i.e. PLUS, can relax these assumptions and incorporate uncertainty in estimating the state and its dynamics in time, and how policies would be different under such analyses.

Our proposed method, PLUS, is a scalable solution to adaptive management of complex ecological systems under both state and model uncertainties. Using simulations, we show that optimal solutions in the literature that ignore measurement error, despite embracing both stochasticity and uncertainty in models and parameters, frequently lead to very poor outcomes in both ecological and economic terms (over-estimating and underestimating scenarios in Fig. 2) and often to complete collapse of the population, since uncertainty about the present observations frustrates learning the correct model (MDP in Fig. 1). Nevertheless, our proposed approach is able to quickly learn the correct model or parameter values that lead to desirable economic and ecological outcomes despite having access to only partial or noisy measurements. These simulation results show the importance of incorporating both model and state uncertainties in the management process and reducing it through adaptive management by learning from arrival of new data.

Furthermore, we illustrate the above conclusions on an application to the management of a real-world marine ecosystem of Argentine hake in Patagonian Shelf (Fig. 3). The actual harvest rules applied in this region resulted in a significant decline in the population biomass from 1985 to 2000, which the population could not recover after that. Here we show that adaptive management through PLUS starts with the harvest level similar to the historical observations; however, with continuing observations of population decline, PLUS adjusts the harvesting policy to level lower than the historical catch, which allows the

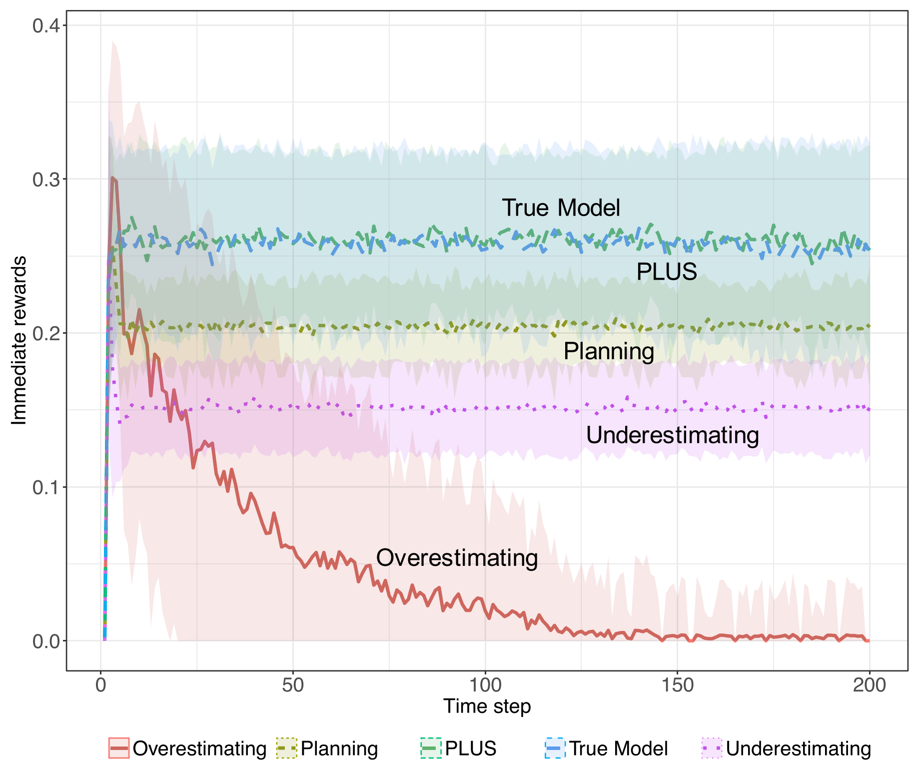

Fig. 2. This figure shows the mean and +/− standard deviation results of 100 independent forward simulations of managing an ecosystem, where population dynamics is governed by Ricker model with uncertainty in the parameter *r*, the intrinsic productivity of the population. Different scenarios in the figure are: (1) PLUS: which integrates adaptive planning and learning proposed here, (2) True Model: perfect knowledge of the true parameter of the model governing the dynamics (which is the upper bound of performance), (3) Planning: management that does not learn from arrival of new data and only finds optimal policies according to maximizing the long-term economic return, by fixing the dynamics model according to historical data, (4) Underestimating: management that underestimates parameter *r* in the dynamics, and (5) Overestimating: agent that overestimates parameter *r*. Y-axis shows the immediate reward (i.e. economic return) that agent receives over time as a feedback from the environment. The measurement error in the process is assumed to be $\sigma_m = 0.5$.

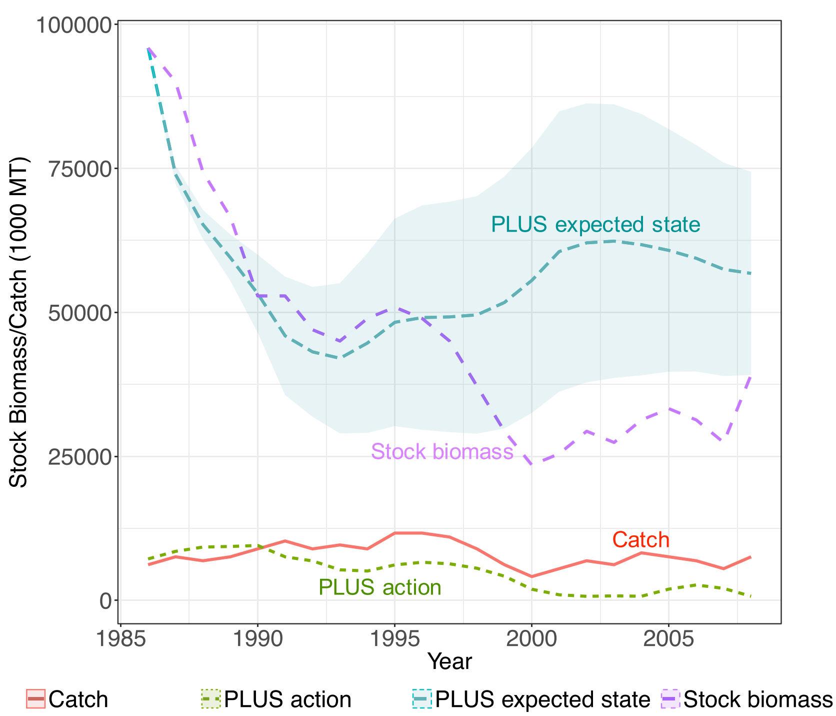

Fig. 3. Performance of the proposed method (i.e. PLUS) on a real-world marine ecosystem of Argentine hake in the Patagonian Shelf in comparison with the historical management. The shaded regions of PLUS expected state shows the mean and +/− standard deviation according to 100 independent simulations of projecting the state trajectory according to PLUS's harvesting decision rule.

population to recover to a sustainable level by the year 2000. It should be noted that the policy suggested by adaptive management through PLUS not only recovers the population but also results in a higher economic return in long-term.

Despite efficiency and scalability of the current implementation of PLUS, it suffers from an assumption that uncertainty exist among a set of candidate models. In cases where the true population model is not among (or adequately approximated by) any of the candidate models, this approach will fail. To simplify the presentation, our examples have assumed that the candidate models all follow the same structural form (e.g. all Ricker or all Beverton-Holt). However, this can be avoided by incorporating more candidate models (which might differ in underlying population model as well as number of parameters) in the prior set. For example, a manager might be interested to learn whether a historical observation of population biomass and harvest can be characterized more accurately by a simple population model such as Beverton-Holt (Beverton and Holt, 1957) or a more complicated age-structure model (Holden and Conrad, 2015) and can include both models with uncertain parameters into the prior set. Hence, there is a need for careful selection of the prior set of candidate models based on the nature of the problem. Generally, non-parametric adaptive management is possible with PLUS, if the manager is completely unsure as to whether the system would follow a specific population model and considering that sufficient amount of data can be gathered. However, it should be noted that such extension is beyond the scope of this manuscript and current available data and is part of our future work. The method we propose here for decision optimization is directly applicable to non-parametric representation of the population model (and/or to a more extreme setting where the dimension of the state space is unknown to the manager; Memarzadeh and Pozzi, 2017). However, the learning cannot be done using the Bayes formula, and approximate methods such as Markov chain Monte Carlo should be adopted (Memarzadeh et al., 2014; Britten et al., 2017).

Moreover, in the current implementation, we have assumed that the population models follow a stationary and time-invariant behavior. However, recent observations have suggested that forces such as climate change and other forms of environmental variations violate this assumption. Although, researchers have tried to relax this assumption before (Pozzi et al., 2017), their proposed approaches are limited to extremely simplified examples and/or fully observable systems. Hence, there is a need for further endeavor of the current implementation to extend the formulations shown here to incorporate the effects of non-stationary dynamics due to climate change feasibly and applicable to real-world complex problems.

An important issue to discuss regarding the implementation of PLUS is its feasibility to be applied in practice. Considering the high computational complexity of adaptive management proposed here, and the need for time-to-time updating of the knowledge on population models and identifying optimal harvesting rules, the practical implementation might seem hard facing real-world political constraints. However, currently 68% of global fisheries are in bad biological and economic conditions (Costello et al., 2016), which shows that the previous practice and implementation need significant revisions. As a result, practitioners need to move towards adaptation of more complex models capable of incorporating several sources of uncertainty and mimicking the real-world complexity.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biocon.2018.05.009.

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